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Innovative Temperature Log Analysis a Field-Scale Framework for Enhanced Well Integrity and Integration

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Abstract

This study introduces a novel, systematic methodology for analyzing temperature logs at a field scale, moving beyond traditional single-well evaluation. By integrating temperature data with geological, geophysical, and operational datasets, the approach enhances well integrity assessment, optimizes workover planning, and supports proactive field-wide integrity management.

The methodology follows a multi-step process: initial downhole temperature logs establish a baseline, while deviations flag potential wellbore integrity issues. Advanced tools such as electromagnetic (EM) and noise logs are deployed to validate anomalies and measure downhole temperature gradients. Cross-well comparisons and spatial correlations with regional geological and hydrogeological features focus on detecting lateral flows and behind-casing cross-flow, providing detailed, actionable insights.

By leveraging temperature mapping and integrating it with other datasets, the approach identifies zones with potential subsurface anomalies and continuity between wells. These patterns inform predictions of integrity risks and highlight the role of well design and cement quality in high-risk areas. A ranking system prioritizes wells for further evaluation, enabling strategic resource allocation and enhancing operational planning.

This methodology shifts the focus from individual wells to a holistic, field-centric integrity evaluation. By aligning thermal anomalies with geological and operational factors, it provides a structured framework for informed intervention planning, supporting proactive well integrity management and improving overall field performance.

Introduction

Well integrity plays a critical role in ensuring that production fluids flow through optimized and safe pathways. Accurate assessment of well integrity status is essential for environmental protection and safeguarding field operator assets.

Several traditional techniques widely employed in the oil and gas industry to achieve downhole sealing include:

- Installation of packers
- Cementation behind casing
- Sealing liner head assemblies
- Setting scab liners

However, due to the inherent complexity of oil and gas wells, completion integrity failures can occur from various factors such as design deficiencies, manufacturing flaws, installation errors, operational challenges, and environmental influences. Common types of completion integrity issues are related to:

- **Tubing and Casing:** These involve damages or breaches in tubing or casing, potentially causing fluid leaks, gas migration, or compromised well control. Causes typically include corrosion, mechanical stresses, improper installation, or inadequate cementing practices.
- **Annular Seal:** Annular seals are essential for maintaining barriers between distinct wellbore zones. Failures of these seals can lead to unwanted fluid migration, loss of zonal isolation, and reduced overall integrity. Typical causes include poor cement jobs, insufficient bonding, or mechanical damage.
- **Equipment Malfunctions:** Completion systems comprise critical equipment such as packers, valves, and safety devices. Malfunctions or failures of these components can severely affect completion integrity. Common causes include mechanical wear, manufacturing defects, inadequate maintenance, or operational errors. Also, erosion and corrosion, debris and scaling also cause problems.
- **Sand Production and Sand Control:** Sand production involves the movement of formation solids into the wellbore, damaging equipment, impairing productivity, and potentially causing sand control system failures. Factors contributing to sand-related issues include inadequate sand control strategies, improper well designs, and challenging reservoir conditions.

To proactively detect and manage these integrity issues, periodic temperature logging under shut-in conditions is a practicable part of annual well integrity monitoring programs. Temperature logs provide crucial insights by identifying temperature anomalies associated with fluid movements behind casing (lateral flow or cross-flow), tubing leaks, gas migration, or zonal communication. Under shut-in condition, thermal equilibration occurs in the wellbore, enabling the identification of subtle deviations from the expected geothermal gradient.

Periodic temperature logging helps detect changes in thermal profiles that may indicate integrity breaches or fluid migration pathways. Regular monitoring enables operators to establish baseline temperature profiles and track trends over time, facilitating early detection of integrity issues. Timely identification and intervention based on temperature logging can prevent significant environmental incidents, enhance operational safety, and reduce repair and remediation costs.

By incorporating regular temperature logging into routine well integrity assessments, operators significantly enhance their capability to maintain wellbore safety, optimize operational efficiency, and ensure long-term asset reliability.

In addition, corrosion logging helps detect and assess metal degradation within tubing and casing, vital for predicting failures and planning preventive measures. Noise logging identifies fluid flow behind casing, tubing leaks, and gas migration by detecting acoustic anomalies. Integrating these methods with temperature data offers a robust diagnostic framework, significantly enhancing the accuracy and reliability of well integrity assessments.

High-Resolution Temperature

High-Resolution Temperature Logging (HRT) is a sophisticated geophysical technique that provides critical insights into both reservoir performance and well integrity. HRT achieves remarkable measurement accuracy by conducting logging during both the downward and upward passes, utilizing sensors with a high resolution of 0.003 °C and rapid response time below one second. This precision significantly reduces the sensitivity of recorded temperature profiles to variations in logging speed, setting HRT apart from traditional temperature logging methods.

To eliminate convective mixing effects caused by tool movements, the temperature sensor is optimally positioned at the bottom of the logging assembly.

Interpretation of HRT data relies on establishing a geothermal baseline—a natural temperature gradient influenced by Earth's internal heat. Deviations from this baseline during extended shut-in periods indicate fluid movements within the reservoir or crossflow between layers. Analysis of temperature anomalies, driven by thermodynamic interactions associated with fluid movements, enables the determination of fluid source locations and flow directions (Figure 1).

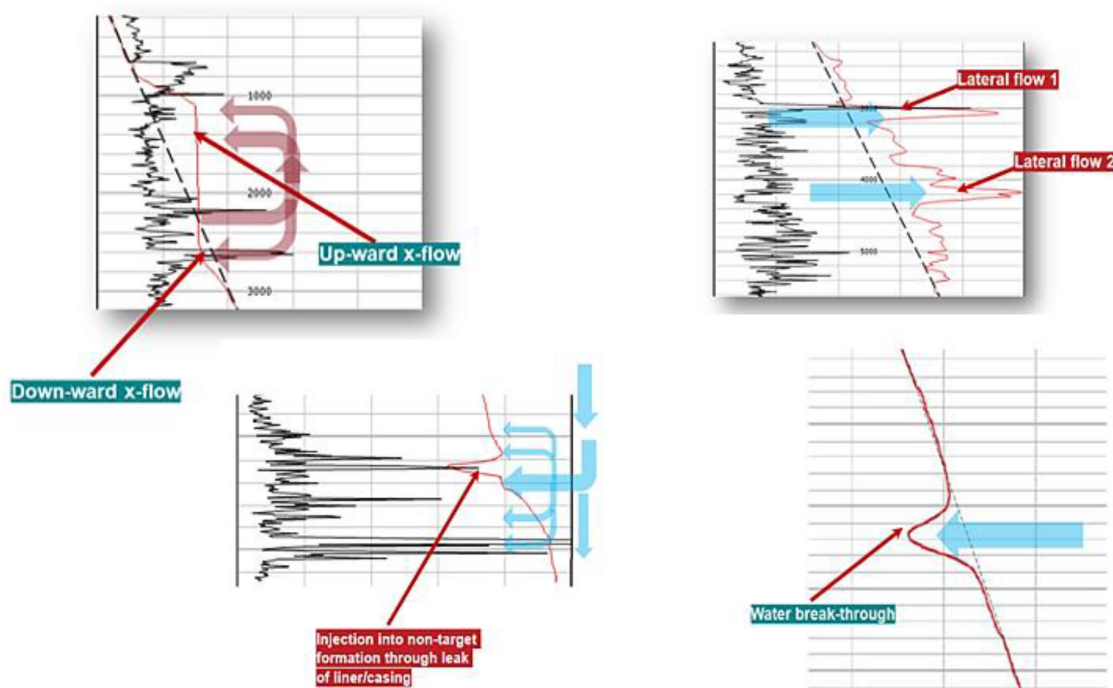


Figure 1—Typical temperature anomalies. Black curve – GR, red curve - temperature

HRT effectively identifies and evaluates three categories of temperature fields:

- Stationary Temperature Fields: Observed in long-term shut-in wells (e.g., observation wells), these fields reflect equilibrium conditions.
- Quasi-Stationary Temperature Fields: These fields occur in wells that have been operating under consistent conditions, resulting from factors such as throttling effects, calorimetric interactions, convective heat transfer, and fluid-rock heat exchanges.
- Transient (Non-Stationary) Temperature Fields: Formed during the recovery of disturbed thermal states resulting from drilling, cementing, perforation, or operational adjustments like well startups, shutdowns, and mode transitions.

Fluid movement through porous media triggers key thermal phenomena such as the Joule-Thomson Effect (JTE). This effect describes the temperature changes fluids undergo when expanding adiabatically from high-pressure areas to low-pressure areas without external work. The Joule-Thomson coefficient ($\mu = \Delta T / \Delta P$) characterizes this temperature shift, varying according to fluid type—positive for liquids like oil and water, negative for gases. Gas expansion typically results in cooling (negative μ), whereas liquids tend to warm due to frictional heating dominating over adiabatic cooling.

Additionally, the calorimetric effect arises when fluids of differing temperatures mix within the wellbore, common in multilayered reservoir environments. The resulting temperature distribution follows a calorimetric equilibrium expressed by:

$$\Delta T_B \times c_B \times Q_B = \Delta T_n \times c_n \times Q_n$$

where ΔT denotes temperature change, c represents specific heat capacity, and Q indicates fluid flow rates. The subscripts "b" and "n" refer respectively to ascending flows from lower layers and inflows from upper layers.

Calorimetric effects are frequently observed near perforations, tubing ends, casing breaches, and during behind-casing crossflows.

Convective heat transfer within the wellbore, annulus, and formation also plays a vital role, fostering temperature equilibrium between flowing fluids and surrounding formations. Although complete thermal equalization is seldom attained, the influence of conductive heat transfer becomes more prominent as flow velocity diminishes. [1]

Corrosion Logging

The Electromagnetic Frequency Domain tool (EMFD) utilizes cutting-edge High-Definition Frequency (HDF) technology to assess well casing conditions. This sophisticated tool works by emitting electromagnetic energy into the surrounding pipes, which then induces eddy currents through multiple concentric well strings. Signals generated at the receiver by the eddy currents in the tubulars are carefully analyzed to provide detailed insights into metal loss and corrosion within the tubulars. The EMFD can evaluate up to seven concentric pipes in a single wellbore, offering a thorough view of the casing's integrity. This allows operators to swiftly identify and address issues such as corrosion-induced defects and metal loss, ensuring the well's structural integrity and safety. The tool's capability to deliver precise, real-time data significantly reduces diagnostic time and boosts the efficiency of wellsite operations.

By utilizing a range of frequencies and multiple spacing receivers, the EMFD tool records extensive signals through multiple barriers allowing the tool to precisely distinguish between various events, including corrosion, defects, bursts, and collapses. The integration of multiple frequencies and receivers facilitates detailed signal analysis and data collection from different investigation depths. This thorough data acquisition process significantly enhances the accuracy and reliability of event differentiation, enabling the identification and analysis of subtle differences between burst and collapse events with greater precision.

Figure 2 depicts the configuration of an EMFD corrosion detection tool used in this study. The tool comprises two transmitting coils and eight receiving coils, designed to operate across a range of frequencies from very low values up to 1000 Hz. It features two distinct sources: one dedicated to high-frequency excitation (High-Resolution, HR) and the other to low-frequency excitation (Primary Array, PA). The receivers are arranged such that the HR source is paired with two short-spaced receivers, while the PA is equipped with six long-spaced receivers.



Figure 2—Frequency-domain EM corrosion detection tool configuration. It shows the EM pipe inspection tool with two arrays of transmitter and receiver pairs. The pair of short-spacing receivers (left section) with high-frequency is employed for inner-most pipe inspections. The pair of long-spacing receivers (right section) with low-frequency is employed for outer-most pipe inspections. Receivers record complex-valued measurements. The acquired data can be then employed to perform evaluation of the multiple pipes.

The characterization of the pipes is accomplished by measuring and analyzing the primary and secondary electromagnetic fields, which interact with the pipes and carry information about their properties. The HR and PA measurements can be processed and displayed as a 2D variable density log (VDL) image. Measurements are normalized with respect to the nominal tool response for the intact pipe. Then different channels of measurement are combined in increasing order of frequency and transmitter-receiver spacing. In this way, the 2D VDL image can clearly indicate the pipe abnormality. For example, abnormality on the innermost pipe affect all the measurements from Rcv1 to Rcv6 – thus showing as wider response in the VDL, while abnormality on outer pipes affect larger-spacing measurements much more pronouncedly – thus showing as narrower response in the VDL. By analyzing the patterns, one can evaluate the pipe events (Figure 3). [2]

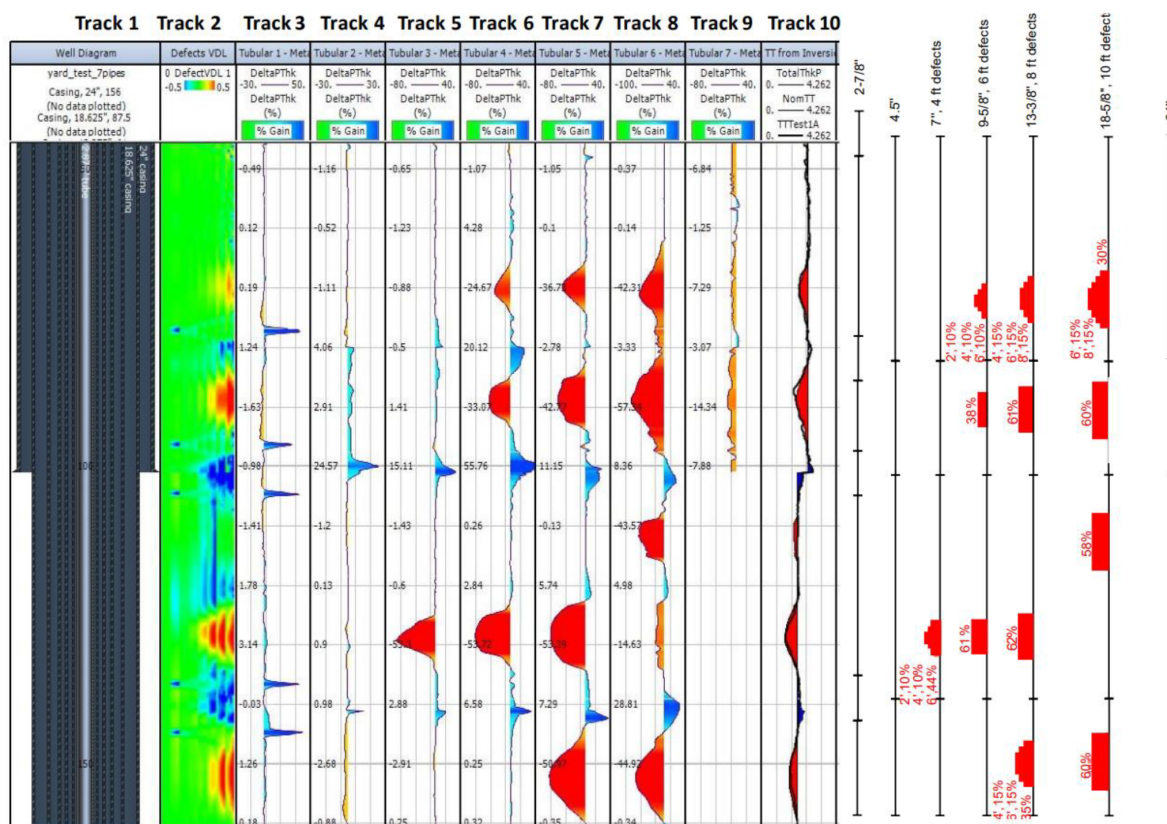


Figure 3—VDL (2nd track) and inversion results of the seven-pipe fixture (Test 2, with cutout on the 7th pipe) including individual pipe metal loss in percentage (from 3rd to 9th tracks) and total thickness (10th track) [3]

Noise Logging

Acoustic Leak Detection Tool (ALDT) is the advanced version of Spectral Noise Logging tool; it is a step-change from previous generations of tools by utilizing an array of hydrophones rather than a typical

single hydrophone. The array consists of eight hydrophones with variable gains which respond to the sound pressure level present in the wellbore. Owing to the array structure, the tools can determine both the vertical and radial location of a noise source and can therefore more precisely indicate which barrier (or barriers) is (are) leaking at a given depth using beamforming triangulation [4].

The Spectral Acoustic Tool is made to record sound in a wide frequency range. The key component of the tool is a highly sensitive hydrophone, which is a piezocrystal sensor or several sensors placed in an oil-filled chamber. The recorded data has a wide frequency range usually from several Hz to up to ~100 kHz. The tool can be deployed by slickline, coiled tubing or wireline with a tractor and can be used in both vertical and horizontal wells. It can provide real-time data with wireline or acquired in memory with a battery pack, which in most cases can power for more than 100 hours. The maximum working wellbore temperature is up to 350 °F and pressures up to 15,000 psi. The tool housing is made of titanium and can be used to log wells containing H₂S and CO₂ (Figure 4).

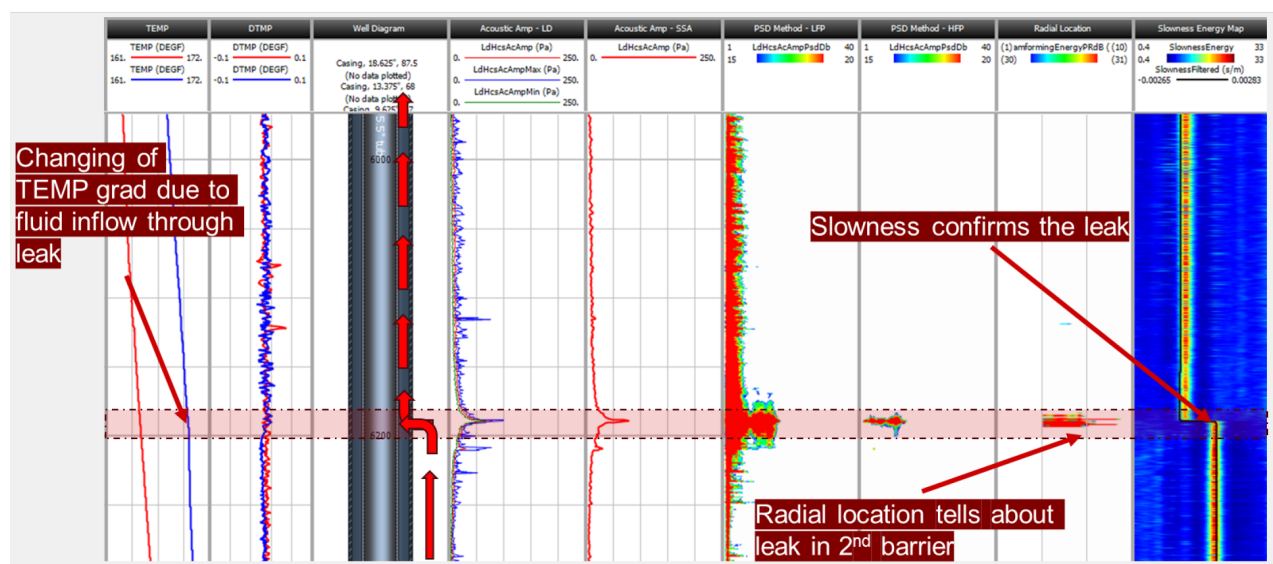


Figure 4—Example of ALDT with radial location determination

Challenge

Today it is common for industry to acquire temperature data during annual shut-in conditions for detecting well integrity issues, yet its full predictive potential remains underutilized.

Temperature data recorded annually under shut-in condition represents a critical and widely used source of information for evaluating the current state of well integrity. This approach involves systematically comparing recent temperature profiles against baseline data, ideally acquired soon after the well was drilled and following a prolonged shut-in period. Such baseline measurements provide an essential reference for accurately detecting and interpreting subsequent deviations or anomalies.

This comparison methodology enables the identification of new thermal events or changes that could signify developing or existing integrity problems. Specific anomalies, such as unexpected temperature gradients, abnormal heat or cooling zones, or shifts in the baseline geothermal gradient, necessitate further investigation. These temperature anomalies could be symptomatic of serious integrity issues, including cement bond degradation, casing corrosion, casing leaks, or compromised zonal isolation with penetration of several barriers inside the well system.

Early detection and detailed analysis of these anomalies through temperature surveys facilitate timely intervention. Operators can initiate additional diagnostics such as electromagnetic logging, noise logging, or cement evaluation logs to confirm and precisely characterize the underlying issues (Figure 5). Consequently,

implementing annual shut-in temperature logging supports proactive well integrity management, reducing risks associated with environmental impacts, maintaining operational safety, and optimizing resource utilization. Ultimately, this rigorous and systematic analysis contributes significantly to ensuring the longevity and reliability of oil and gas production assets.

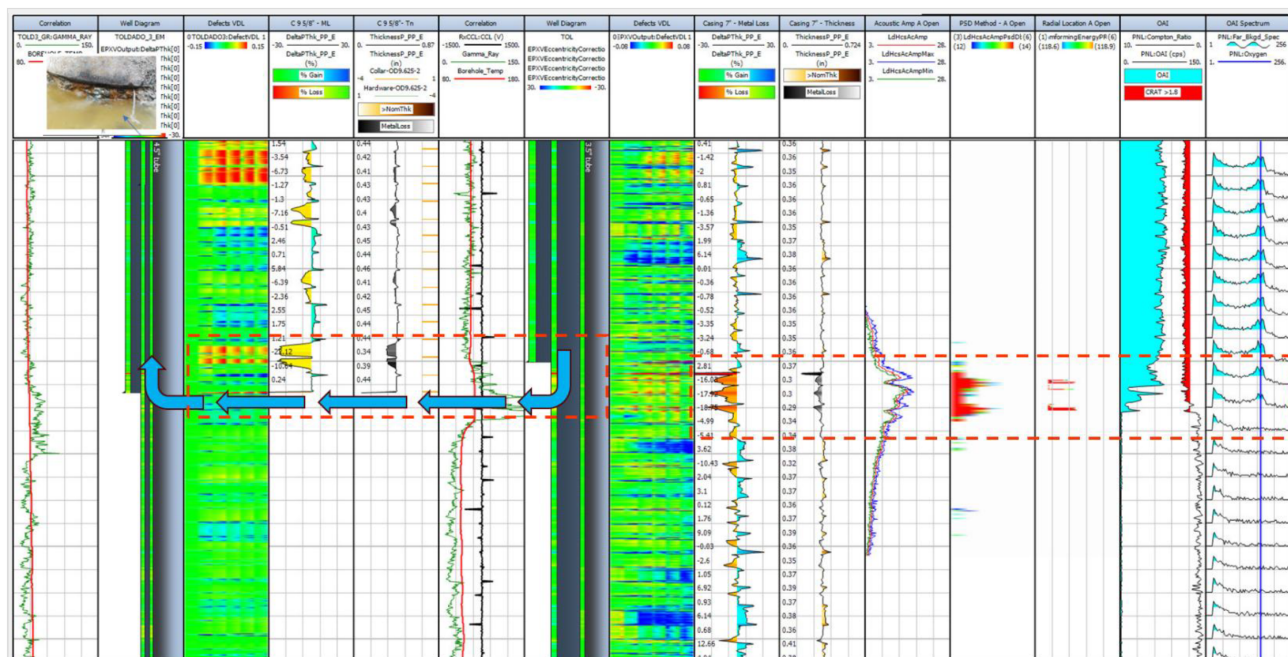


Figure 5—Example of combination temperature, EM and ADLT data for leak detection

However, what if operators could proactively mitigate well integrity issues before a planned well is even drilled? Leveraging existing temperature data from one well can significantly inform and improve the design and planning of subsequent wells. If a temperature anomaly is identified in a reference well, there is a high probability that the same anomaly or similar event may occur in a new well with a comparable construction design, especially when drilling through identical geological formations, including aggressive aquifers, and targeting the same production zone.

This predictive capability stems from the consistency of large-scale environmental and operational conditions. In areas characterized by stable system behavior, regional patterns in pressure regimes and fluid movement are typically sustained over wide areas. Consequently, thermal and pressure-related phenomena observed in one location are likely to appear in adjacent wells under comparable operational scenarios.

For example, in regions with extensive, interconnected reservoir systems, lateral fluid movement originating from distant zones can significantly affect local conditions. One notable mechanism is the Infiltration Water Pressure System (IWPS), in which infiltrated surface water creates pressure gradients that drive lateral fluid flow within subsurface systems (Figure 6). These systems generally involve:

- Recharge zones where surface water enters the subsurface.
- Discharge zones where water exits into adjacent aquifers or formations or at the surface (springs, wetlands).

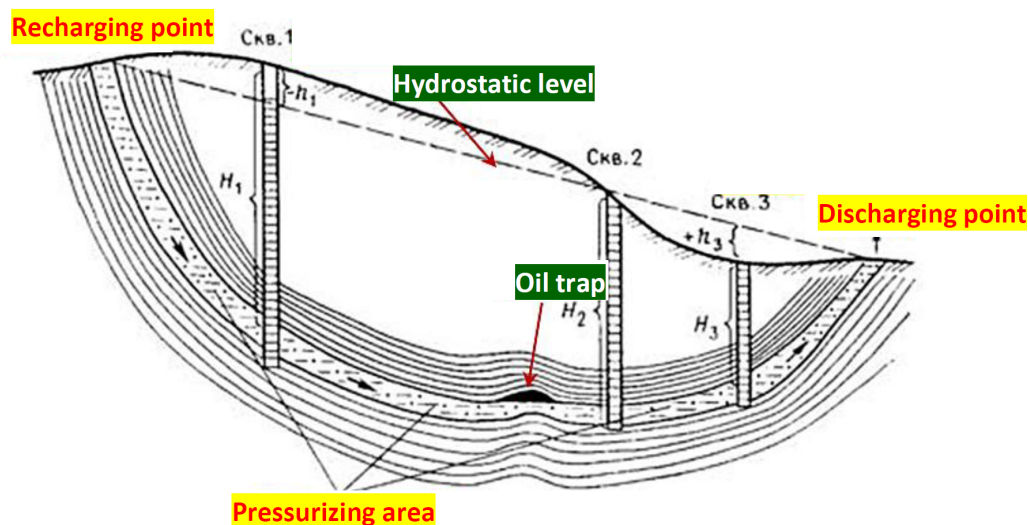


Figure 6—Infiltration water pressure system example

Within such systems, sustained lateral flow can impact operational performance by:

- Inducing inter-zonal fluid migration, lateral flows and crossflows.
- Transmitting pressure across well formations.
- Accelerating corrosion due to fluid interactions or cement degradation behind casing, and,
- Producing temperature anomalies through thermal exchange between reservoir fluids and infiltrated waters.

Understanding these mechanisms allows operators not only to detect and respond to existing integrity threats but also to anticipate and mitigate potential risks in future developments. By incorporating temperature analysis into both monitoring and design strategies, operators can enhance asset reliability, reduce intervention costs, and optimize field development planning.

Solution

Conventional temperature logs with stationary measurements every 100 or 500 ft in single well, especially when the HRT sensor is not utilized, cannot reveal the full hydrodynamic situation around the well system. It can help to suggest the current situation. But the correlation of the HRT temperature log with Noise logging, EMRD and CBL data is critical to revealing the suspected cause of the integrity issue.

Meanwhile, if review of the neighbor well that has the same design, and the temperature log shows similar fluid movement, there is a high chance that an integrity issue is already there or it's starting to propagate and in short time will create the problem for that well. (Figure 7).

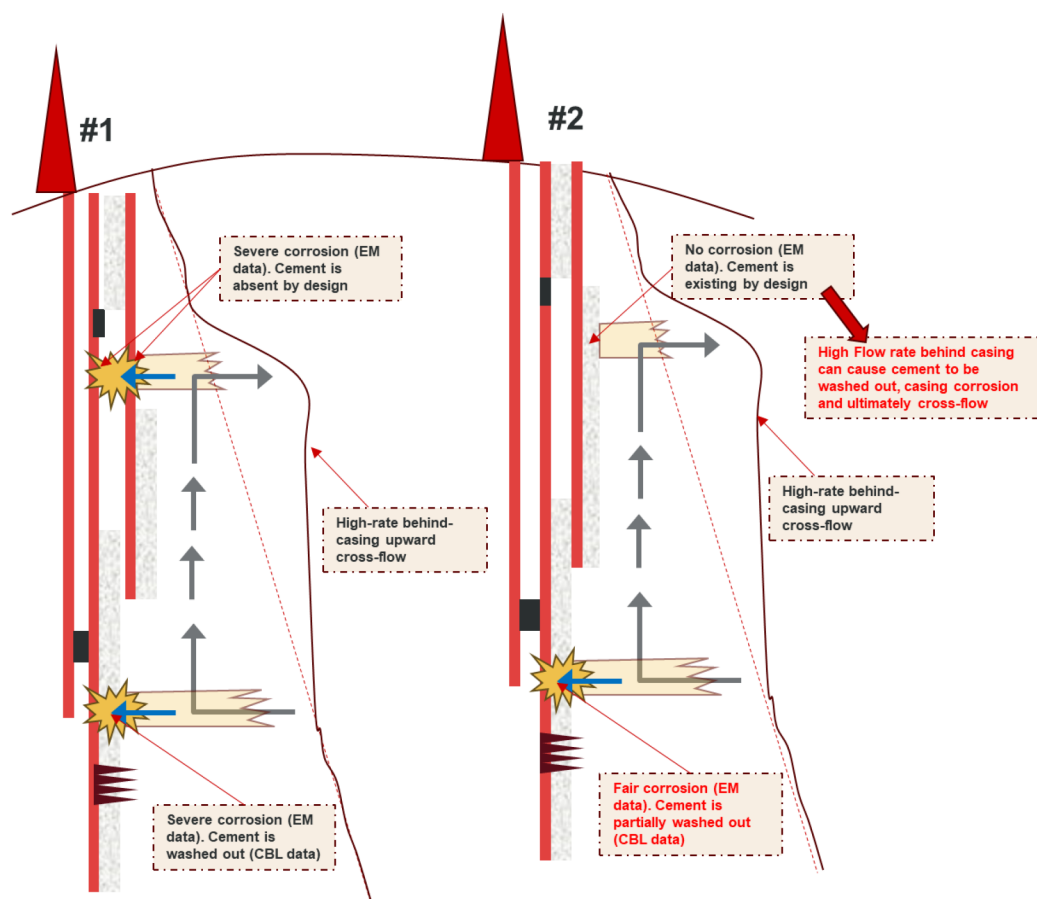


Figure 7—Suspected area hydrodynamic based on temperature anomalies

Analysis of the shut-in temperature logs for the whole field (instead of looking at one or two wells) can provide more clarifications regarding the integrity status of the zone of interest. In this manuscript, the objective is to present how detailed knowledge of temperature anomalies and geological phenomena observed in existing wells—at both field and regional scales—can be systematically applied to inform the design, construction, and operational strategies of future wells. By leveraging these insights, operators can proactively mitigate the risk of encountering the same integrity issues currently observed in producing wells. Ultimately, this approach not only enhances operational safety and optimizes well design but also reduces environmental impacts and improves the long-term reliability and performance of hydrocarbon assets.

Conducting temperature analyses across an entire field, rather than focusing solely on individual wells, provides a comprehensive and holistic understanding of the reservoir dynamics and overall well integrity. A field-wide approach enables operators to detect and interpret regional patterns, anomalies, and trends that might not be visible when evaluating isolated well data.

Analyzing temperature data from multiple wells throughout the field offers several critical benefits:

- Enhanced Detection of Regional Trends and Anomalies.** By examining temperature profiles across multiple wells simultaneously, operators can quickly identify regional thermal anomalies such as lateral flows, cross-zone fluid migration, or cooling anomalies due to water break-through or injection to unwanted areas or even leaks of the casing in injection wells. Early identification of these areal and regional anomalies enables proactive interventions, reducing the risk of widespread integrity issues and ensuring continuous operational efficiency.
- Improved Predictive Capabilities and Preventive Measures.** Field-wide temperature analyses enhance the predictive understanding of integrity risks. Recognizing temperature anomalies

consistently occurring in neighboring wells allows operators to anticipate potential issues in adjacent wells, proactively deploying preventive maintenance measures. This approach significantly reduces operational risks and potential downtime caused by unforeseen integrity failures.

- **Optimized Workover and Maintenance Planning.** Comprehensive temperature analysis across a field allows prioritization of interventions based on the severity and urgency of anomalies identified. Operators can allocate resources strategically and efficiently, targeting maintenance efforts on wells demonstrating the highest risks, thereby optimizing resource use and reducing unnecessary expenditures.
- **Integration with Geological and Hydrogeological Data.** Integrating field-wide temperature data with geological, seismic, hydrogeological, and lithological datasets greatly enhances the accuracy of integrity assessments. Cross-validation with external geological data facilitates more robust interpretation of temperature anomalies, improving diagnostic precision. This integrated perspective helps clarify the underlying geological and operational causes of anomalies, guiding more informed and effective intervention strategies.
- **Long-Term Asset Sustainability and Operational Safety.** By adopting a comprehensive field-wide temperature analysis approach, operators enhance overall operational safety, proactively manage reservoir performance, and ensure the sustainability of asset integrity over the long term. Regular and systematic analyses facilitate continuous improvement in understanding reservoir dynamics, allowing for timely corrective actions and ongoing optimization of field development strategies.
- **Enhanced Well Casing Design and Metallurgy Selection.** Insights gained from temperature anomalies and thermal stress patterns allow operators to optimize well casing design and select appropriate casing metallurgy. This ensures that materials and well architecture are properly matched to the anticipated thermal, chemical, and mechanical conditions, reducing the risk of future integrity failures and extending the operational lifespan of the wells.

Therefore, field-wide temperature analysis significantly strengthens the overall management of well integrity and reservoir performance, providing a strategic advantage through improved operational safety, optimized resource allocation, and enhanced long-term asset management.

Methodology

A comprehensive approach to temperature logging across an entire field involves several methodical steps, from initial data acquisition to strategic recommendations for intervention.

Step 1: Field-wide Temperature Data Acquisition and Analysis

Temperature logs from all available wells are systematically collected and analyzed. Historical data, pre-production baseline temperatures, and current temperature surveys are integrated into thermal gradient maps, revealing regions of concern. Thermal modeling of subsurface conditions aids in interpreting these data, highlighting anomalies indicating potential integrity issues.

Step 2: Identification of Temperature Patterns and Anomalies

Thermal profiles are compared against established geothermal baselines to identify significant deviations or anomalies. Key types of anomalies captured during shut-in condition include (Figure 1 and 8):

- Hot cone-shape anomalies typical for lateral flows – no penetration into borehole but affecting the cement bond and corroding outer casing barriers.

- Same results but for cool cone-shape anomalies that are caused by water breakthrough from neighbor water injector. But also, injection into non-target interval through the leak of casing from inner to outer area is creating similar cooling anomaly due to cooling down the initial reservoir temperature – "residual cooling anomaly".
- Cooling anomalies due to downward cross-flows between two points. It can be a sign of leakage in metal barriers of well system and also, about activity behind-casing that leads to cement bond and metal casing damage.
- Similar results to well integrity if the heating anomalies due to upward cross-flows between two points were detected.

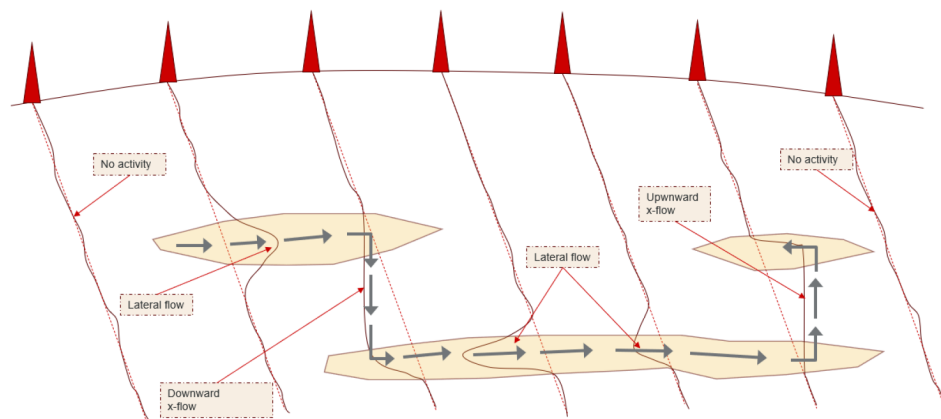


Figure 8—Suspected area hydrodynamic based on temperature anomalies

Step 3: Ranking Wells Based on Identified Anomalies

Each well is systematically categorized into ranks based on anomaly severity and the likelihood of integrity issues (Figure 9):

- **Rank 1:** Wells displaying normal temperature gradients without indications of integrity problems – baseline for area/field temperature analyzes.
- **Rank 2:** Wells with lateral flows and water break-through anomalies suggesting possible integrity issues behind the casing without penetration into borehole. Targets for cement quality evaluation and corrosion logging. But for injectors such cooling anomalies can be a sign of leak in casings from inner to outer area.
- **Rank 3:** Wells showing downward and /or upward cross-flow anomalies indicating high possibility of integrity issues appearing. Target wells for noise and corrosion logging, production logging and cement evaluation data will add value.

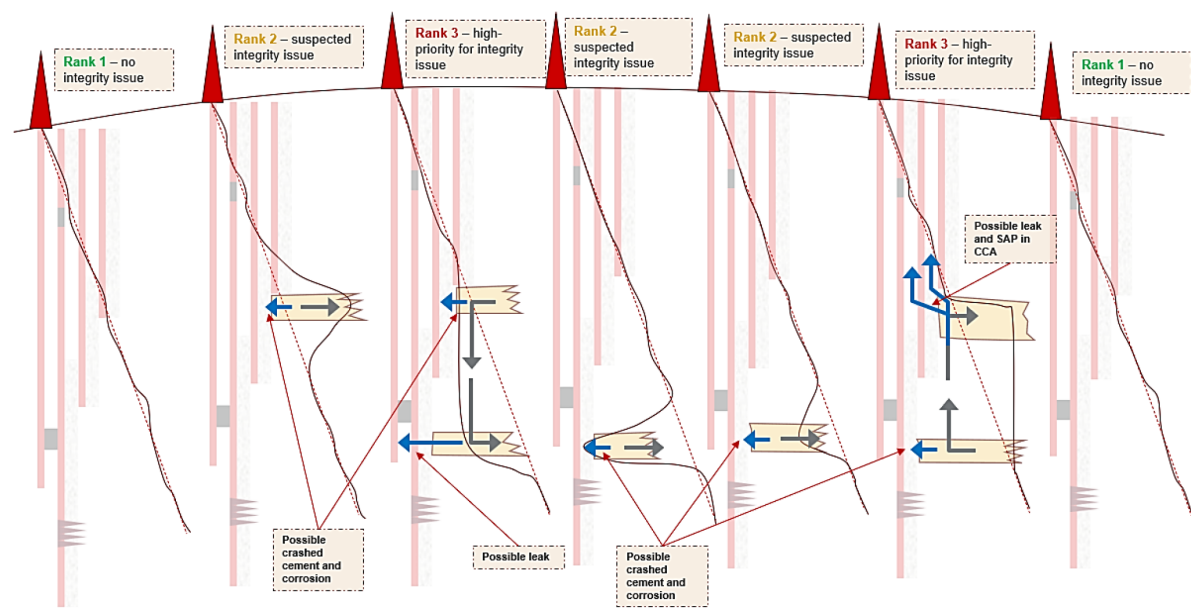


Figure 9—Suspected area hydrodynamic based on temperature anomalies

Step 4: Targeted Recommendations for Further Investigations

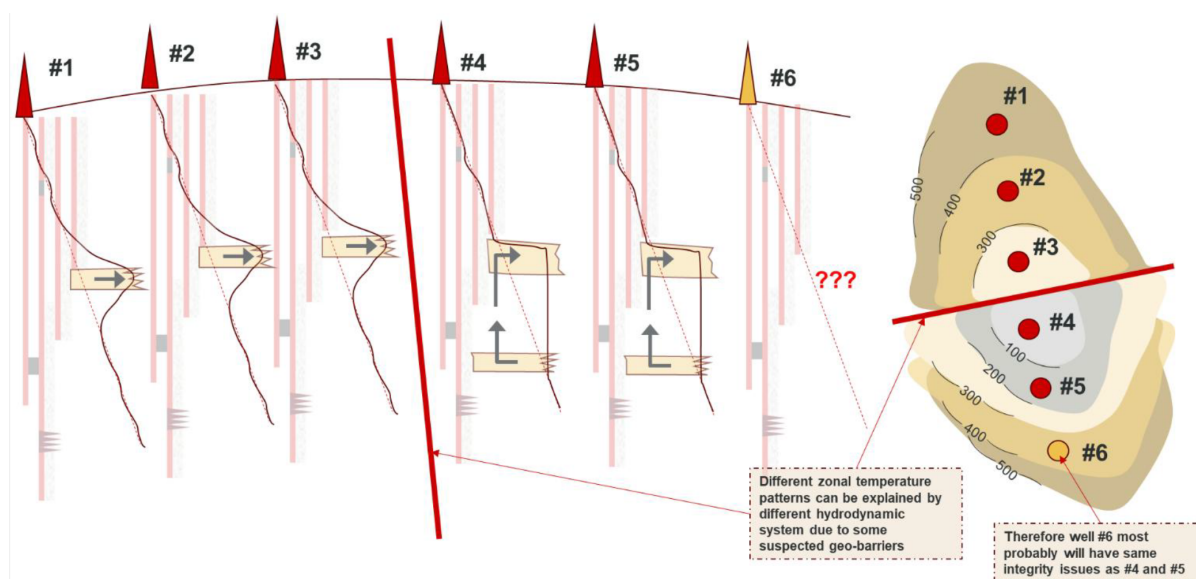
Specific investigative recommendations are tailored based on the ranking:

- **Rank 1:** Routine annual temperature surveys to maintain surveillance.
- **Rank 2:** Further detailed assessment using time-lapse Electromagnetic (EMFD) and temperature logging. Time-lapse techniques see the propagation of metal loss and temperature anomalies in time domain. It helps to make preventive actions before the situation goes to Rank 3 class. It is recommended to monitor surface annuli pressures to react to surface annular pressure (SAP) changes on time.
- **Rank 3:** Comprehensive examination by noise logging – full leak detection job with possible PLT and EM tools support. This integrated approach precisely identifies and characterizes integrity issues, facilitating immediate remediation actions.

Step 5: Integration with Regional Geological Data

Temperature anomalies are further analyzed in conjunction with geological, hydrogeological, seismic, and lithological datasets. Cross-validation with these external factors provides additional clarity and insight into the causes of temperature deviations, enhancing predictive capabilities and informing strategic intervention and workover planning.

Mapping temperature events and correlation with structural geological map of aquifers can explain the different hydrodynamic systems between two zones (wells ##1-3 and wells ##4-5) which is most probably caused by geological barriers. Therefore, drilling any new wells with the same design and materials of construction over those two zones, there is a high probability that it will have similar integrity issues as neighbor wells (Figure 10).



In this case, there is a good motivation to update the well or cement design to stop the influence of the flow activity to the well sealing system and prevent the integrity issue from occurring.

Step 6: Strategic Recommendations for Clients

Clients receive actionable recommendations based on the confirmed ranked analysis after EMFD and Noise logging surveys:

- **Rank 1 Wells:** Continuous monitoring and maintenance scheduling to maintain integrity.
- **Rank 2 Wells:** Planned EMFD logging, preventive maintenance if possible, and further evaluation to preemptively address potential issues.
- **Rank 3 Wells:** Urgent detailed evaluation using integrated logging techniques (EMFD and Noise), followed by immediate workover or remediation planning to resolve identified integrity issues promptly.

By implementing this structured approach, operators can optimize field-wide integrity management, strategically plan interventions, effectively allocate resources, and improve long-term reservoir and operational efficiency. This comprehensive methodology ensures robust management of reservoir and well integrity, reducing operational risks and enhancing asset sustainability.

Conclusions

This study demonstrates the significant advantages of applying field-wide temperature log analysis as a proactive well integrity management tool. By shifting the focus from single-well evaluation to a field-centric approach, operators can detect regional patterns, identify early-stage integrity risks, and optimize the prioritization of well interventions.

The integration of high-resolution temperature data with noise logging, electromagnetic surveys, and geological datasets provides a comprehensive understanding of subsurface conditions, allowing for precise identification of lateral flows, crossflows, and behind-casing anomalies. The proposed ranking methodology enables systematic categorization of wells, facilitating targeted diagnostic and remedial actions.

Importantly, the insights gained from temperature anomalies and hydrodynamic patterns directly inform enhancements in well design and the selection of appropriate casing metallurgy, reducing future integrity risks and extending the operational life of assets. This integrated approach not only strengthens

operational safety and environmental stewardship but also improves resource allocation and long-term asset sustainability.

Overall, the presented methodology offers a robust and scalable framework for improving well integrity management across complex fields, ultimately contributing to safer, more efficient, and environmentally responsible operations.

Abbreviations

HRT	- High-Resolution Temperature Logging;
JTE	- Joule-Thomson Effect;
EMFD	- Electromagnetic Frequency Domain Tool;
IWPS	- Infiltration Water Pressure System;
CBL	- Cement Bond Logging;
PLT	- Production Logging Tool
SAP	- Sustained Annulus Pressure;
ALDT	- Acoustic Leak Detection Tool

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